

BiDexAffordance: Learning Collaborative Affordances for Efficient Bimanual Dexterous Grasping

Authors omitted for double-anonymous review

Abstract—Synthesizing bimanual dexterous grasps requires selecting compatible contact regions while optimizing high-dimensional hand configurations under collision and stability constraints. We present BiDexAffordance, which learns collaborative surface-contact priors from simulation-validated bimanual grasps to initialize physical refinement. A first-hand heatmap proposes a contact region, and a second heatmap conditioned on the selected contact predicts a complementary region for the other hand. The resulting pair initializes wrist poses before a coupled optimizer refines both wrists and finger configurations. On unseen instances from an 11-category benchmark containing 170 objects, BiDexAffordance achieves 35.9% macro-average grasp success, compared with 13.9% for random-initialized BimanGrasp, using 1,000 versus 10,000 refinement iterations. On a physical dual-arm platform, it succeeds in 42 of 90 executed lifting trials, compared with 24 of 90 for BimanGrasp under the same candidate-selection and reachability-filtering protocol. These results support collaborative contact prediction as an effective initialization strategy for bimanual dexterous grasp synthesis.

I. INTRODUCTION

Bimanual dexterous grasping is an important capability for robots handling wide, bulky, or awkwardly shaped objects. Two hands can establish separated contact regions to distribute load and resist rotation, as illustrated in Fig. 1. Dexterous hands further offer multiple contact configurations around these regions. Realizing this capability requires coordinating wrist poses and finger articulation while respecting object geometry, hand-hand collision, and grasp stability.

The resulting synthesis problem combines a high-dimensional configuration space with strong inter-hand coupling. Physics-based methods such as BimanGrasp [1] can optimize grasps for new objects, but random initialization often places the hands in unfavorable regions and leads to long searches or failed refinement. Learning a direct distribution over both hand poses amortizes this search, yet must simultaneously represent contact selection and detailed articulated configurations. An object-centric contact representation separates these two levels: learning proposes where the hands should cooperate, and optimization determines how they realize those contacts.

A useful initialization must identify contact regions that work together: two regions that each support a plausible single-hand grasp may still yield an incompatible pair. We therefore introduce BiDexAffordance, which first predicts a surface-contact heatmap for one hand and then predicts a complementary region conditioned on the selected first contact. The resulting pair initializes wrist placement and approach directions before a coupled physical optimizer refines both hands.

Our design builds on contact-based grasp representations [2]–[4] and conditional affordance prediction for two grippers [5]. The challenge here is to make this decomposition useful for two dexterous robot hands: a surface contact leaves many wrist orientations and finger configurations unresolved. We learn coarse collaborative regions from validated dual-hand grasps and delegate detailed contact formation to refinement. The affordance maps act at initialization, while the optimizer retains its collision and stability terms.

We evaluate the resulting pipeline on a benchmark of 170 objects from 11 categories, two additional held-out categories, and a physical dual-arm system. On unseen instances, success increases from 13.9% to 35.9% with 1,000 versus 10,000 BimanGrasp refinement iterations. The ablations assess how contact conditioning and physical refinement affect grasp success. Additional comparisons examine heatmap quality, supervision source, and reference-contact initialization.

Our contributions are threefold. First, we formulate contact initialization as sequential prediction of a first-hand heatmap and a conditional second-hand heatmap, and connect the predicted pair to geometric wrist initialization for bimanual dexterous refinement. Second, we construct paired heatmap supervision automatically from physically validated dual-hand grasps. Third, we evaluate grasp success across object instances and two held-out categories, analyze contact conditioning and optimizer coupling, and validate the resulting grasps in physical lifting trials. Efficiency here refers to refinement iterations; end-to-end wall-clock throughput also includes perception, inference, and validation.

II. RELATED WORK

A. Bimanual Dexterous Grasping and Manipulation

DexGraspNet [6] and BimanGrasp [1] synthesize robotic grasps through optimization and physical validation. We build on this physical foundation and learn a contact prior that improves initialization. BimArt [7] synthesizes bimanual human-hand interactions with articulated objects, while ManipTrans [8] transfers dexterous bimanual manipulation through residual learning. More recent systems explore text-conditioned dual-hand grasping (DHAGrasp [9]), synthesis and policy learning across bimanual strategies (UltraDexGrasp [10]), and synthetic human pre-grasps with robot retargeting and refinement (SynManDex [11]). Our focus is the contribution of collaborative surface-contact initialization to final grasp synthesis within a fixed optimizer.

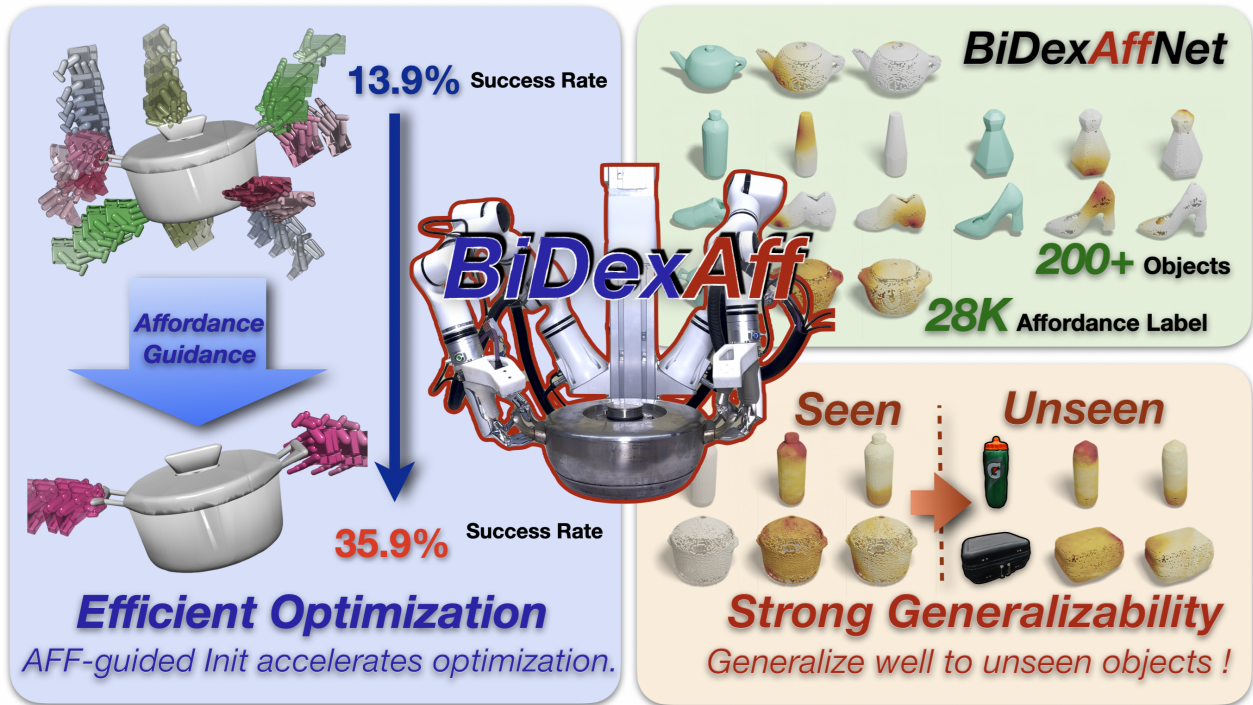


Fig. 1. BiDexAffordance learns collaborative contact priors from validated bimanual grasps to initialize physical refinement. On unseen instances, BimanGrasp achieves 13.9% success with 10,000 refinement iterations; Ours (Full) achieves 35.9% with 1,000 (Table I). These budgets measure refinement iterations. BiDexAffNet denotes the supervision dataset; the benchmark evaluated here comprises 170 objects.

Teleoperation and imitation-learning systems such as ALOHA [12] and Diffusion Policy [13] address trajectory generation and feedback control, complementing our focus on terminal-grasp initialization.

B. Contact Representations and Collaborative Affordances

GenDexGrasp [2] uses contact maps as an intermediate representation for robotic dexterous grasping. Contact2Grasp [3] decomposes generation into contact prediction and pose synthesis, and ContactGen [4] augments contact location with hand-part and direction information for human grasp synthesis. ContactOpt [14] predicts desired contact and refines an input hand pose to match it. In comparison, we predict one coarse region per hand to initialize a coupled robotic optimizer; detailed finger contacts emerge during physical refinement.

DualAfford [5] establishes conditional affordance prediction for dual-gripper manipulation using affordance, proposal, and critic modules. Building on this decomposition, our predictors take an object point cloud and the selected first contact, learn paired surface heatmaps from validated dual-hand grasps, and convert the predicted contacts into wrist initializations for articulated physical refinement. Paired heatmaps provide direct supervision for predictor training; articulated grasp optimization is performed during data generation and final refinement.

C. Affordance Supervision from Human and Robot Data

GRAB [15], ARCTIC [16], OAKINK2 [17], and AffordPose [18] capture rich human contact and functional

interactions. Robot-specific synthesis instead aligns labels with a particular embodiment and physical holding objective. Our dataset construction derives paired contact heatmaps from validated bimanual robot grasps on a curated object set. The human-versus-robot supervision experiment examines the practical effect of this choice; task intent, contact coverage, and morphology can all influence transfer.

III. PRELIMINARIES

A. Problem Formulation

Given an object surface point set $O = \{x_i\}_{i=1}^N$, we seek a bimanual grasp

$$G = \{(R_1, t_1, \theta_1), (R_2, t_2, \theta_2)\}, \quad (1)$$

where $R_h \in SO(3)$ and $t_h \in \mathbb{R}^3$ specify the wrist pose of hand h , and θ_h contains its joint angles. The grasp should retain the rigid object while satisfying collision and joint constraints. The predictors operate on $N = 8192$ sampled surface points; the downstream optimizer additionally uses object and hand geometry for distance and penetration evaluation.

B. Collaborative Bimanual Affordance Maps

We represent a contact prior using a surface heatmap $A_h \in [0, 1]^N$. The first-hand map $A_1(x_i; O)$ scores candidate regions for the first-hand contact. After selecting $k_1 \in O$, the conditional map $A_2(x_i; O, k_1)$ scores complementary regions

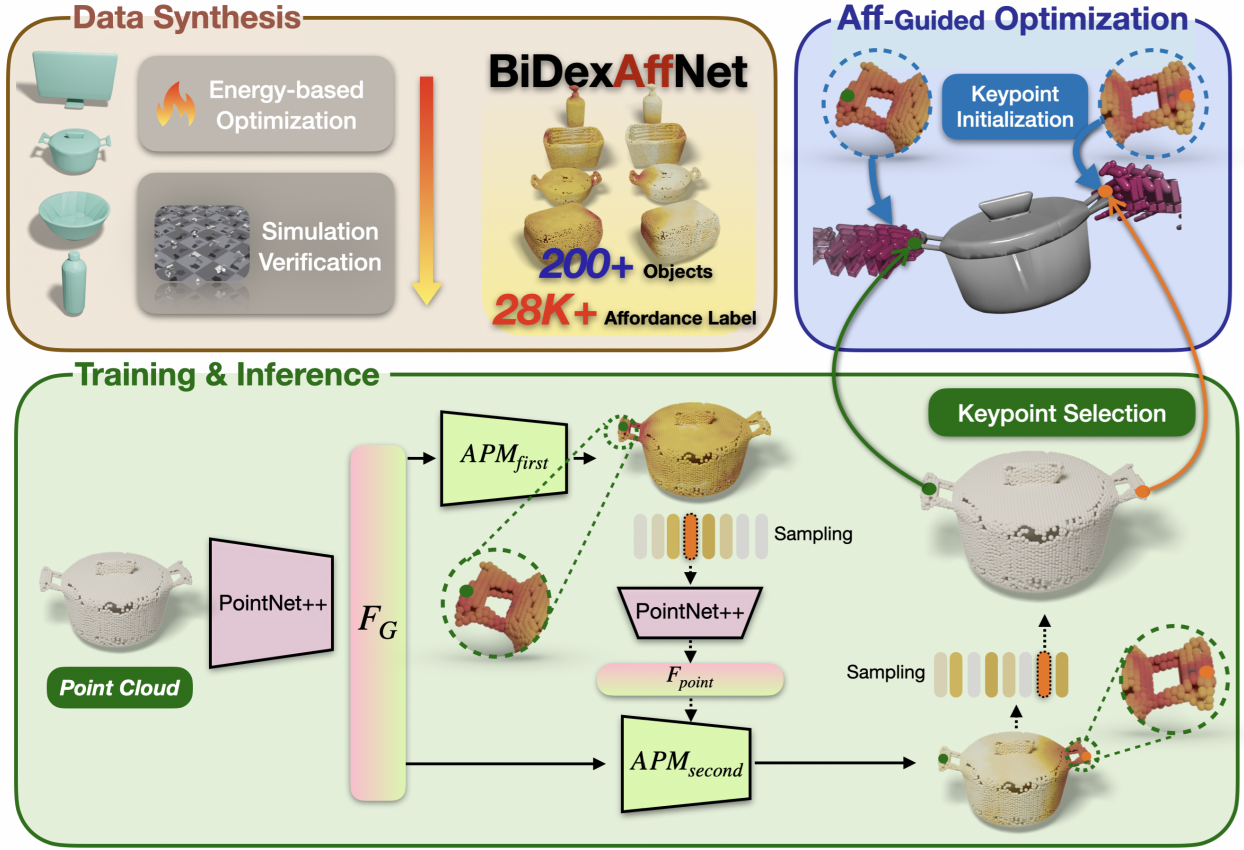


Fig. 2. Overview of BiDexAffordance: validated grasp synthesis, sequential contact prediction, and initialization for physical refinement. Training conditions the second predictor on the paired ground-truth first contact; inference uses the selected predicted first contact. Each keypoint summarizes a hand-level contact region and initializes a wrist pose before full-hand refinement.

for the second hand. Normalizing the scores for sampling defines

$$p(k_1, k_2 | O) = p_1(k_1 | O) p_2(k_2 | O, k_1). \quad (2)$$

This construction samples contact pairs through two surface predictions without storing a dense $N \times N$ pair table. It preserves the full wrist-and-joint state for subsequent optimization. First and second denote prediction order, independent of the robot’s execution order.

In this work, an *affordance map* summarizes regions associated with successful rigid-object holding; its role in synthesis is a *contact prior*. The scores are supervised heatmap values rather than calibrated success probabilities. Task semantics and temporal manipulation policies lie outside this representation.

IV. METHOD

Figure 2 summarizes the pipeline. We first synthesize and validate dual-hand grasps, extract paired contact supervision, and train the two affordance predictors. At inference, predicted contacts provide geometric initialization for physical refinement.

A. Automatic Contact Supervision

Grasp generation and validation. We generate candidates with BimanGrasp [1] and retain grasps passing Isaac

Gym [19] validation under the criterion in Sec. V-A. The resulting collection, BiDexAffNet, contains approximately 28K validated poses on 170 objects. Converting these poses into surface heatmaps makes their cooperative contact structure available as direct supervision. Label extraction uses geometry and validated poses, without manually annotating affordances on each object.

Contact extraction. For each accepted grasp, we transform 4,096 sampled surface points of each hand into the grasp pose. Let $\mathcal{F}_h$ be these points and D_O the object axis-aligned bounding-box diagonal. We identify nearby object points by

$$d_{ih} = \min_{f \in \mathcal{F}_h} \|x_i - f\|_2, \quad \mathcal{N}_h = \{i : d_{ih} \leq 0.02D_O\}. \quad (3)$$

When $\mathcal{N}_h$ is nonempty, let $w_{ih} = 1/(d_{ih} + \epsilon)$. We compute an inverse-distance-weighted centroid and project it to the sampled surface:

$$\begin{aligned} \tilde{k}_h &= \frac{\sum_{i \in \mathcal{N}_h} w_{ih} x_i}{\sum_{i \in \mathcal{N}_h} w_{ih}}, \\ k_h &= \arg \min_{x_i \in O} \|x_i - \tilde{k}_h\|_2. \end{aligned} \quad (4)$$

Here $\epsilon > 0$ ensures numerical stability. If the proximity set is empty, we use the object point closest to the hand. Each keypoint summarizes a hand-level contact region.

Heatmap targets. For an extracted keypoint k_h , the target is

$$Y_h(x_i; k_h) = \exp\left(-\frac{\|x_i - k_h\|_2^2}{2s^2}\right), \quad (5)$$

where s is a bandwidth in the coordinates of O , distinct from the proximity radius $0.02D_O$. Let $\mathcal{G}_O$ denote the validated grasps used to construct supervision for object O . The first predictor regresses the object-level target

$$\bar{Y}_1^O(x_i) = \frac{\sum_{g \in \mathcal{G}_O} Y_1(x_i; k_1^g)}{\max_j \sum_{g \in \mathcal{G}_O} Y_1(x_j; k_1^g)}. \quad (6)$$

For the second predictor, each example retains the first contact k_1^g and the second-hand target $Y_2(x_i; k_2^g)$ from the same validated grasp g . This pairing preserves the cooperative structure of the supervision.

B. Collaborative Affordance Prediction

Both predictors use PointNet++ [20] set abstraction and feature propagation, with the same backbone design and separate parameters. Feature propagation recovers a 128-dimensional representation at each input point, followed by a pointwise prediction head and sigmoid activation.

The first predictor takes object coordinates. The second receives the first-contact encoding

$$r_i = x_i - k_1, \quad \phi_i = [r_i; \|r_i\|_2], \quad (7)$$

which provides both direction and distance relative to the selected contact. These features enter the conditional encoder and its feature-propagation path. The regression objectives distinguish the aggregated first-hand target from the paired second-hand target:

$$\begin{aligned} \mathcal{L}_1^O &= \frac{1}{N} \sum_{i=1}^N |A_1(x_i; O) - \bar{Y}_1^O(x_i)|, \\ \mathcal{L}_2^g &= \frac{1}{N} \sum_{i=1}^N |A_2(x_i; O, k_1^g) - Y_2(x_i; k_2^g)|. \end{aligned} \quad (8)$$

The second predictor uses ground-truth first contacts during training and predicted contacts at inference. The two heatmaps supply the scores for sequential contact selection. Selecting the first contact breaks object symmetry and allows an asymmetric, complementary second-hand map. Table III reports heatmap quality under the two conditioning settings.

C. Affordance-Guided Initialization and Refinement

Geometric initialization. For selected contact k_h and object centroid c_O , we compute an inward approach direction

$$a_h = \frac{c_O - k_h}{\|c_O - k_h\|_2}. \quad (9)$$

The wrist rotation aligns the palm normal with a_h , leaving a twist angle about this axis to specify the remaining orientation. With approach offset $d > 0$ and palm-center displacement Δp_h in the wrist frame, we initialize

$$p_h = k_h - da_h, \quad t_h = p_h - R_h \Delta p_h. \quad (10)$$

Thus, approach directions are computed from geometry rather than adjusted manually per object. This provides a starting configuration near the predicted contact region. The centroid direction is a heuristic; collision resolution remains part of refinement, particularly for concave objects.

Physical refinement. Starting from these wrist poses and initial joint configurations, we optimize the BimanGrasp energy

$$\begin{aligned} E(G) &= \lambda_{\text{dis}} E_{\text{dis}} + \lambda_{\text{fc}} E_{\text{fc}} + \lambda_{\text{pen}} (E_{\text{obj}} + E_{\text{self}}) \\ &\quad + \lambda_{\text{bim}} E_{\text{bim}} + \lambda_{\text{joint}} E_{\text{joint}}. \end{aligned} \quad (11)$$

The terms account for hand-object distance, a force-closure surrogate, object penetration, self-penetration, inter-hand penetration, and joint limits. We retain the coupled physical objective: the contact maps initialize the grasp but do not enter the energy as additional constraints. Contact locations can move during refinement, and final success is determined by physical validation. Optimizer hyperparameters are shared across object categories.

V. EXPERIMENTS

We evaluate grasp success, generalization, and refinement efficiency, then analyze the learned affordances and validate the pipeline on a physical dual-arm system.

A. Experimental Setup

Dataset. The benchmark comprises 170 objects across 11 categories and approximately 28K validated bimanual poses. Objects are drawn from SAPIEN-related assets [21]–[23] and BimanGrasp [1]. We use ShadowHand models in simulation and evaluate seen and unseen object instances within each category. Lamps and baskets provide an additional held-out-category test.

Physical evaluation. For each object, we generate 1,024 candidates and validate them in Isaac Gym using the DexGraspNet-style protocol [6]. Each grasp is tested under gravity along six axis-aligned directions for 100 simulation steps. Success requires retaining the object under all directions with maximum penetration no greater than 0.1 cm. The reported aggregation first computes the fraction of successful candidates for each object and then averages these fractions within each category. We recompute the macro averages in Table I as the equally weighted mean of the 11 displayed category rates. This last calculation uses the rounded table values, not recovered trial-level records. The metric counts individual candidates, without dual-arm reachability filtering.

Baselines and ablations. Uni2Bim independently refines two hands with a single-hand DexGraspNet optimizer for 10,000 refinement iterations. BimanGrasp uses coupled optimization from random initialization for 10,000 iterations. Our DDPM+Opt baseline learns a final-pose prior from synthetic grasps using a denoising diffusion model [24], followed by 1,000 BimanGrasp refinement iterations. *Ours (Full)* denotes BiDexAffordance, also with 1,000 refinement iterations. We evaluate two variants: *Ours (Two-map)* predicts two unconditioned surface heatmaps, and *Ours (Uni)* uses conditional contact initialization with independent

TABLE I
GRASP SUCCESS ON SEEN AND UNSEEN OBJECT INSTANCES (%).

Category	Baselines						Ours					
	Uni2Bim		BimanGrasp		DDPM+Opt		Two-map		Uni		Full	
	Seen	Unseen	Seen	Unseen	Seen	Unseen	Seen	Unseen	Seen	Unseen	Seen	Unseen
Bottles	2.9	2.9	10.7	7.4	13.1	11.0	12.5	19.8	27.1	27.4	33.8	32.4
Bowls	4.5	4.4	9.5	10.6	16.2	9.8	18.8	18.7	32.9	39.6	34.2	39.2
Clocks	2.7	3.9	10.4	9.6	9.3	8.7	9.2	14.7	40.2	49.2	44.2	53.2
Containers	6.5	5.2	15.2	15.1	13.2	12.4	27.7	36.7	42.6	39.3	44.3	39.2
Dispensers	3.8	3.4	10.1	8.5	5.8	3.3	9.4	11.6	39.3	31.9	36.9	30.4
Displays	9.1	8.6	29.9	30.6	23.8	25.5	24.9	24.2	38.2	40.0	40.5	37.6
Kettles	2.8	2.3	3.3	3.1	5.8	3.3	6.1	5.7	35.6	33.7	31.2	34.8
Plates	5.3	5.7	22.6	21.8	23.8	25.5	18.1	15.0	39.6	32.9	43.0	38.1
Pots	0.6	0.3	0.9	0.5	1.3	0.3	8.4	7.7	11.5	7.7	11.1	8.1
Shoes	4.5	3.1	21.9	20.5	35.7	23.1	20.2	15.5	49.3	51.6	49.3	51.8
Toys	9.3	7.0	25.0	25.4	24.0	21.5	21.5	24.0	36.0	27.9	36.8	30.1
Macro avg.	4.7	4.3	14.5	13.9	15.6	13.1	16.1	17.6	35.7	34.7	36.8	35.9
Iterations	10,000		10,000		1,000		1,000		1,000		1,000	

Under Ours, Two-map and Uni denote the ablations and Full denotes BiDexAffordance. Bold marks the best result in each split. Aggregation is defined in Sec. V-A; iterations count physical refinement.

single-hand refinement. Both use 1,000 iterations. The Two-map implementation has one backbone with two output channels, whereas the conditional variant uses two separately parameterized predictors. All methods follow the same physical success criterion.

B. Grasp Success and Instance Generalization

Table I shows that BiDexAffordance achieves 35.9% macro-average success on unseen instances, compared with 13.9% for BimanGrasp, a gain of 22.0 percentage points using one tenth of the refinement iterations. It also exceeds DDPM+Opt under the same 1,000-iteration refinement budget. Improvements over BimanGrasp occur in all 11 unseen categories, supporting the use of learned object-surface contacts for initialization.

The seen and unseen macro averages are 36.8% and 35.9%, respectively, indicating transfer to new instances within the evaluated categories. Performance remains category dependent: unseen success ranges from 8.1% on pots to 53.2% on clocks. The learned prior improves the search for cooperative configurations, while geometry-dependent difficulties remain in articulated refinement.

Contact conditioning and coupled refinement. At 1,000 refinement iterations, the conditional variant achieves 35.9% unseen-instance success, compared with 17.6% for the implemented Two-map variant, a difference of 18.3 percentage points. With conditional initialization, coupled refinement improves the macro average from 34.7% to 35.9%, although independent refinement performs better in four categories: bowls, containers, dispensers, and displays. These comparisons support collaborative initialization as a useful component of the pipeline, with a smaller aggregate gain from coupled refinement. Because the predictor architectures

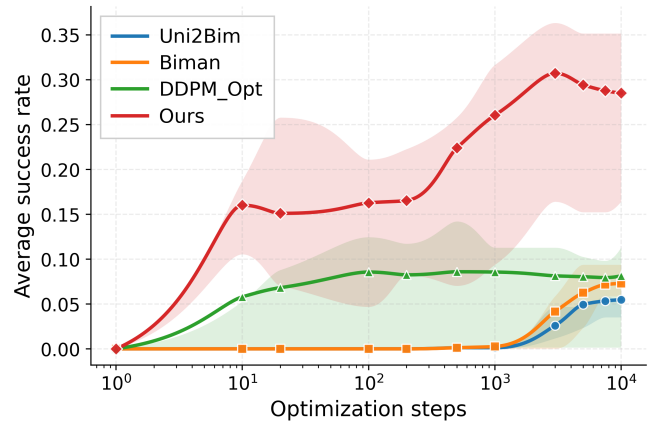


Fig. 3. Success versus refinement iterations on 11 unseen objects, one per category. Biman and DDPM_Opt denote BimanGrasp and DDPM+Opt. The comparison measures iteration budgets; shaded bands are not interpreted as confidence intervals.

differ, the Two-map comparison does not isolate conditioning at matched capacity.

With affordance initialization, the independent-refinement variant reaches 34.7% unseen-instance success at 1,000 iterations, compared with 4.3% for Uni2Bim at 10,000 iterations. This comparison involves both initialization and refinement-budget differences.

C. Refinement Budget

Table I provides two budget comparisons. At 1,000 refinement iterations, contact initialization yields 35.9% unseen-instance success, compared with 13.1% for DDPM+Opt. Relative to random-initialized BimanGrasp, it achieves higher success with one tenth of the refinement iterations. These

TABLE II
SUCCESS ON CATEGORIES EXCLUDED FROM TRAINING (%).

Method	Lamps	Baskets
Uni2Bim	3.2	5.1
BimanGrasp	14.8	18.6
DDPM+Opt	11.3	15.9
Ours (Two-map)	18.9	21.4
Ours (Uni)	32.7	35.2
Ours (Full)	35.8	39.5

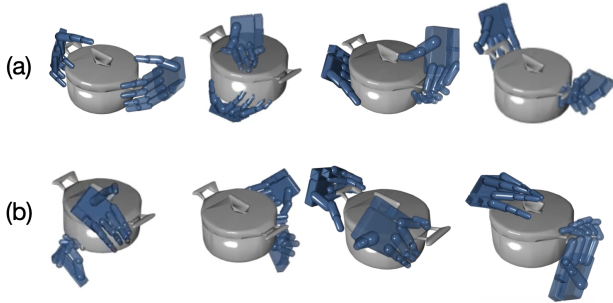


Fig. 4. Qualitative pose diversity: four generated grasps from (a) BiDex-Affordance and (b) BimanGrasp on one object, showing different refined wrist and finger configurations.

results concern the reported budgets; end-to-end runtime also depends on perception, network inference, per-iteration computation, and validation. Figure 3 presents the reported iteration-budget comparison on a separate subset of 11 unseen objects, one per category. The main quantitative budget comparisons above use Table I.

D. Generalization to Unseen Categories

Table II reports results on lamps and baskets, which are excluded from the training categories. BiDexAffordance obtains 35.8% and 39.5%, compared with 14.8% and 18.6% for BimanGrasp. Conditional prediction also improves over the Two-map variant in both categories. These results extend the instance-level evaluation to the two evaluated held-out categories.

E. Affordance Prediction Quality

We evaluate heatmaps against the surface labels used for supervision. For average precision (AP), target scores of at least $\tau = 0.5$ define positive points; predicted scores provide the ranking. IoU@Top- k measures the intersection over union of the highest-scoring $k\%$ of points in the predicted and target heatmaps, for $k \in \{1, 5\}$. For the second hand, we compare ground-truth first-contact conditioning with predicted first-contact conditioning.

The first-hand map reaches 0.71 AP. The second-hand predictor reaches 0.63 AP with ground-truth conditioning and 0.48 with predicted conditioning; the corresponding top-5% IoU changes from 0.69 to 0.61. The reported scores are lower under predicted conditioning. Interpreting this difference requires accounting for how the second-hand target

TABLE III
AFFORDANCE PREDICTION QUALITY.

Map	First contact	AP	IoU@1%	IoU@5%
First	–	0.71	0.59	0.62
Second	Ground truth	0.63	0.64	0.69
Second	Predicted	0.48	0.52	0.61

IoU@ $k\%$ uses the top $k\%$ of points in each heatmap.

TABLE IV
SUPERVISION-SOURCE COMPARISON ON 10 ARCTIC OBJECTS.

Supervision	Success (%)
Human affordance	39.7
Robot-synthesized affordance	58.2

is matched to the selected first contact. Downstream grasp success remains the principal measure of the full pipeline.

F. Human and Robot Supervision

On a separate set of 10 ARCTIC objects, we compare a model trained with human interaction supervision with one trained using robot-synthesized supervision, followed by the same downstream optimization pipeline. Table IV reports 39.7% and 58.2% success, respectively. Robot-synthesized supervision yields higher success on this evaluation. The difference may reflect task, contact-coverage, and embodiment differences between the supervision sources. We treat this experiment as a supervision-source comparison and keep its results separate from the main benchmark.

G. Reference-Contact Initialization

We replace predicted keypoints with reference contacts sampled from ground-truth affordance labels while retaining the same optimizer and 1,000-refinement-iteration budget. This replacement improves success by 1.5 percentage points, from 35.9% to 37.4% (Table V). The result characterizes this initialization protocol, not a formal upper bound: wrist orientations, initial joint configurations, label coverage, and physical refinement still affect the grasp.

H. Pose Diversity and Failure Modes

Figure 4 compares four grasps per method, illustrating different refined wrist and finger arrangements on the same object. These examples show variation among the generated solutions; distributional coverage is not measured by this qualitative comparison.

Qualitative failure inspection identifies three recurring mechanisms: unsuitable predicted contacts, penetration during refinement, and collision-free configurations that fail to retain the object under gravity. These qualitative failure modes can overlap.

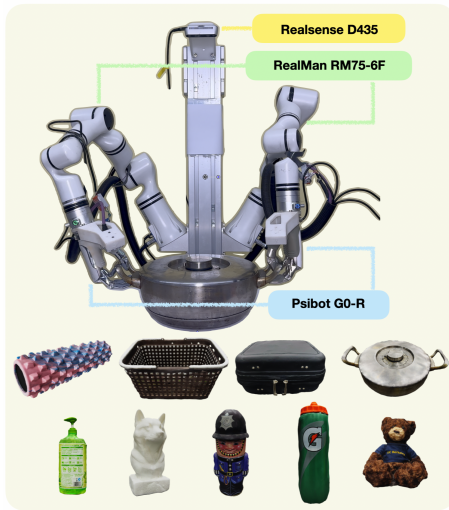


Fig. 5. Physical platform and nine test objects: two RealMan RM75-6F arms, Psibot G0-R hands, and a RealSense D435 camera.

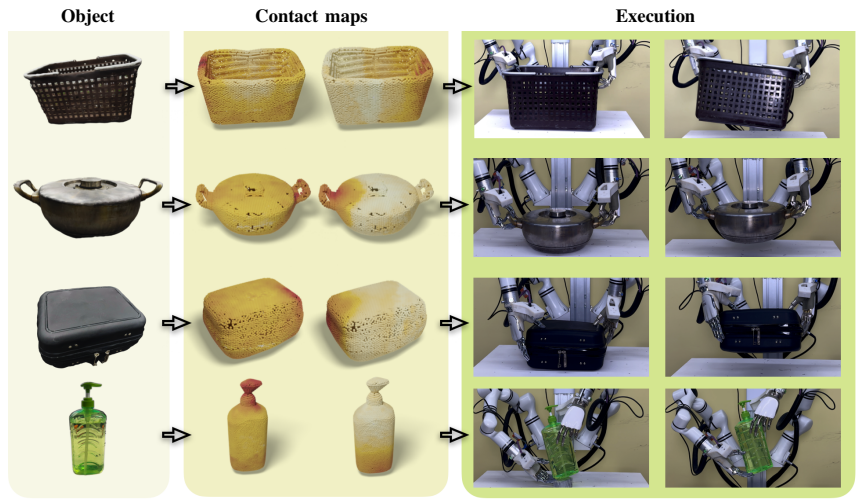


Fig. 6. Object geometry, predicted contact maps, and executed grasps on four example objects. Within each row, the two maps correspond to the first and second hands, and the execution images show grasping and lifting. Red regions indicate higher predicted contact scores. Table VI evaluates all nine objects.

TABLE V
CONTACT INITIALIZATION ON UNSEEN INSTANCES.

Initialization	Iterations	Success (%)
Random (BimanGrasp)	10,000	13.9
Predicted contacts	1,000	35.9
Reference contacts (GT labels)	1,000	37.4

TABLE VI
REAL-ROBOT LIFTING ON NINE OBJECTS.

Method	Successes / trials	Rate (%)
BimanGrasp	24 / 90	26.7
DDPM+Opt	22 / 90	24.4
Ours (Full)	42 / 90	46.7

Ten executed trials per object and method, after candidate selection and reachability filtering.

I. Real-World Experiments

Setup and execution. The platform comprises two RealMan RM75-6F arms, Psibot G0-R hands, and a RealSense D435 camera (Fig. 5). We calibrate the camera to the robot with an ArUco marker and acquire multi-view object observations to reconstruct geometry. From this geometry, we sample 8,192 surface points and run affordance prediction followed by grasp refinement. Grasp poses are transformed into the robot frame and screened for dual-arm reachability before execution. The robot moves to a pre-grasp pose, approaches the object, and lifts it.

Protocol and results. We test nine objects with ten executed trials per object and method. A successful trial lifts the object by 10 cm and retains it without dropping or substantial slipping. Figure 6 shows contact predictions and executions.

Among 90 executed trials per method, BiDexAffordance succeeds in 42 trials (46.7%), compared with 24 (26.7%) for BimanGrasp and 22 (24.4%) for DDPM+Opt (Table VI). These rates are conditional on candidate selection and reachability filtering.

Rejected candidates are outside the 90 executed trials. Simulation instead evaluates an unfiltered candidate pool under six gravity directions, with a different hand embodiment. The real and simulation rates therefore characterize different tests. Within the shared execution protocol, the lifting results support the practical benefit of the learned initialization.

VI. DISCUSSION AND LIMITATIONS

The strongest evidence concerns collaborative contact initialization: the conditional pipeline outperforms the implemented Two-map variant, and conditional initialization also improves success with independent refinement. Coupling the optimizers gives a smaller aggregate improvement, with gains that vary by category. Absolute success remains modest, including on pots, so contact prediction and articulated refinement remain complementary. The reference-contact experiment measures one substitution protocol without identifying a unique cause for the remaining failures.

The supervision inherits the sampling preferences and physical assumptions of the grasp generator, and the second predictor encounters ground-truth contacts during training but predicted contacts at inference. Our evaluation concerns rigid-object holding in the tested instances and categories. Functional tasks require task-specific contacts and trajectories, and the current tests do not establish that every object requires two dexterous hands. Physical execution relies on reconstructed multi-view geometry, embodiment mapping, candidate selection, and reachability filtering. Reconstruction error and full-arm feasibility therefore remain relevant limits. Capacity-matched predictors, geometric contact heuristics,

controlled adaptations of contact-guided baselines, other optimizers or real contact labels, and end-to-end timing would help isolate the sources and scope of the observed gains.

VII. CONCLUSION

BiDexAffordance learns paired surface-contact priors from validated bimanual grasps and uses them to initialize physical refinement. Conditional contact prediction supports grasp synthesis across the evaluated instances and two held-out categories. In the main simulation benchmark, the method improves unseen-instance success from 13.9% to 35.9% with 1,000 versus 10,000 refinement iterations. Physical lifting trials provide additional evidence under a reachability-filtered execution protocol. Future work will integrate task-conditioned contacts and full-arm feasibility into grasp synthesis.

REFERENCES

- [1] Y. Shao and C. Xiao, “Bimanual grasp synthesis for dexterous robot hands,” *IEEE Robotics and Automation Letters*, vol. 9, no. 12, pp. 11 377–11 384, 2024.
- [2] P. Li, T. Liu, Y. Li, Y. Geng, Y. Zhu, Y. Yang, and S. Huang, “GenDexGrasp: Generalizable dexterous grasping,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2023.
- [3] H. Li, X. Lin, Y. Zhou, X. Li, Y. Huo, J. Chen, and Q. Ye, “Contact2Grasp: 3D grasp synthesis via hand-object contact constraint,” in *Proc. Int. Joint Conf. Artificial Intelligence (IJCAI)*, 2023, pp. 1053–1061.
- [4] S. Liu, Y. Zhou, J. Yang, S. Gupta, and S. Wang, “ContactGen: Generative contact modeling for grasp generation,” in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023.
- [5] Y. Zhao, R. Wu, Z. Chen, Y. Zhang, Q. Fan, K. Mo, and H. Dong, “DualAfford: Learning collaborative visual affordance for dual-gripper manipulation,” in *Int. Conf. Learning Representations (ICLR)*, 2023.
- [6] R. Wang, J. Zhang, J. Chen, Y. Xu, P. Li, T. Liu, and H. Wang, “DexGraspNet: A large-scale robotic dexterous grasp dataset for general objects based on simulation,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2023.
- [7] W. Zhang, R. Dabral, V. Golyanik, V. Choutas, E. Alvarado, T. Beeler, M. Habermann, and C. Theobalt, “BimArt: A unified approach for the synthesis of 3D bimanual interaction with articulated objects,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, June 2025, pp. 27 694–27 705.
- [8] K. Li, P. Li, T. Liu, Y. Li, and S. Huang, “ManipTrans: Efficient dexterous bimanual manipulation transfer via residual learning,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, June 2025, pp. 6991–7003.
- [9] Q. Li, Z. Wu, J. Wang, C. C. Loy, and B. Dai, “DHAGrasp: Synthesizing affordance-aware dual-hand grasps with text instructions,” *arXiv preprint arXiv:2509.22175*, 2025.
- [10] S. Yang, Y. Xie, Z. Liang, Y. Tian, J. Zeng, D. Lin, and J. Pang, “UltraDexGrasp: Learning universal dexterous grasping for bimanual robots with synthetic data,” *arXiv preprint arXiv:2603.05312*, 2026.
- [11] Y. Shao, Z. Chen, W. Lin, M. Zhou, T. Chen, X. Yang, Y. Chi, and Y. Mu, “SynManDex: Synthesizing human-like dexterous grasps from synthetic human pre-grasps,” *arXiv preprint arXiv:2606.09798*, 2026.
- [12] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, “Learning fine-grained bimanual manipulation with low-cost hardware,” in *Robotics: Science and Systems (RSS)*, 2023.
- [13] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, “Diffusion policy: Visuomotor policy learning via action diffusion,” in *Robotics: Science and Systems (RSS)*, 2023.
- [14] P. Grady, C. Tang, C. D. Twigg, M. Vo, S. Brahmbhatt, and C. C. Kemp, “ContactOpt: Optimizing contact to improve grasps,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 1471–1481.
- [15] O. Taheri, N. Ghorbani, M. J. Black, and D. Tzionas, “GRAB: A dataset of whole-body human grasping of objects,” in *European Conference on Computer Vision (ECCV)*, 2020. [Online]. Available: <https://grab.is.tue.mpg.de>
- [16] Z. Fan, O. Taheri, D. Tzionas, M. Kocabas, M. Kaufmann, M. J. Black, and O. Hilliges, “ARCTIC: A dataset for dexterous bimanual hand-object manipulation,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, June 2023, pp. 12 943–12 954.
- [17] X. Zhan, L. Yang, Y. Zhao, K. Mao, H. Xu, Z. Lin, K. Li, and C. Lu, “OAKINK2: A dataset of bimanual hands-object manipulation in complex task completion,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, June 2024, pp. 445–456.
- [18] J. Jian, X. Liu, M. Li, R. Hu, and J. Liu, “AffordPose: A large-scale dataset of hand-object interactions with affordance-driven hand pose,” in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, October 2023, pp. 14 713–14 724.
- [19] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, *et al.*, “Isaac Gym: High performance GPU-based physics simulation for robot learning,” *arXiv preprint arXiv:2108.10470*, 2021.
- [20] C. R. Qi, L. Yi, H. Su, and L. J. Guibas, “PointNet++: Deep hierarchical feature learning on point sets in a metric space,” in *Advances in Neural Information Processing Systems*, 2017.
- [21] F. Xiang, Y. Qin, K. Mo, Y. Xia, H. Zhu, F. Liu, M. Liu, H. Jiang, Y. Yuan, H. Wang, L. Yi, A. X. Chang, L. J. Guibas, and H. Su, “SAPIEN: A simulated part-based interactive environment,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [22] K. Mo, S. Zhu, A. X. Chang, L. Yi, S. Tripathi, L. J. Guibas, and H. Su, “PartNet: A large-scale benchmark for fine-grained and hierarchical part-level 3D object understanding,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019.
- [23] A. X. Chang, T. Funkhouser, L. Guibas, P. Hanrahan, Q. Huang, Z. Li, S. Savarese, M. Savva, S. Song, H. Su, *et al.*, “ShapeNet: An information-rich 3D model repository,” *arXiv preprint arXiv:1512.03012*, 2015.
- [24] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Advances in Neural Information Processing Systems*, 2020. [Online]. Available: <https://arxiv.org/abs/2006.11239>